ORGANIC CHEMISTRY
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Cite this: DOI: 10.1039/c5qo00164a

A concise construction of carbohydrate-tethered axially chiral allenes *via* copper catalysis†

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Received 20th May 2015,
Accepted 15th June 2015

DOI: 10.1039/c5qo00164a

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The efficient construction of the axial chirality of allenes bearing a carbohydrate unit is challenging. Here we observed that the newly developed $CuBr_2$ catalysis may smoothly catalyze the highly diastereoselective three-component reaction of carbohydrates bearing a terminal alkyne unit, aliphatic or aromatic aldehydes, and (*R*) or (*S*)-α,α-diphenylprolinol, affording carbohydrates bearing the chiral 1,3-substituted allene entity in very high de (94–99%). Synthetically attractive functionalities, such as hydroxyl, amide, malonate, and even tosylate, *etc*. could all be well tolerated. The role of the carbohydrate unit has been explored.

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It is well known that chirality is a very common and attractive phenomenon existing in nature. For example, the left hand is a non-superimposable mirror image of the right hand. At the molecular level, organic molecules may contain central, axial, planar, and helical chiralities, *etc*., among which central chirality is the most frequently observed and closely related to the biological activity. Thus, the enantioselective construction of central chirality has been extensively explored. Allenes are a class of compounds with an axial chirality of their two cumulated C=C bonds.[1] In comparison, the axial chirality of allenes has received much less attention and is far from being well established.[2,3] On the other hand, some natural or unnatural carbohydrates bearing allene units show very interesting biological activities, and the allene unit was even found in some marketed drugs (Fig. 1).[4] Considering the biopotential of both carbohydrates and allenes as well as the utility of the chiral allene unit for construction of other units with central chirality *via* chirality transfer, the development of carbohydrates bearing a chiral allene unit is of high interest and importance.

We chose 2-propynyl 2,3,4,6-tetra-*O*-acetyl-β-D-glucopyranoside **1a**[5] as the model substrate. The recently developed ZnX_2-mediated one-pot synthesis of optically active 1,3-disubstituted allenes from terminal alkynes with aldehydes and α,α-diphenylprolinol is straightforward.[6,7] However, initial experiments showed that the reaction under our previously developed protocols with 0.75 equiv. of $ZnBr_2$,[6] or even CdI_2,[8] both failed to afford the expected product (R_a)-**4aa** (Scheme 1). Quite unexpectedly, we observed that the reaction using newly developed $CuBr_2$ catalysis[9] could solve this problem, affording the carbohydrates bearing an allene unit in decent yields with excellent diastereomeric excess!

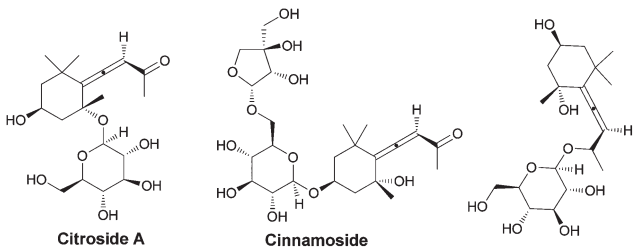
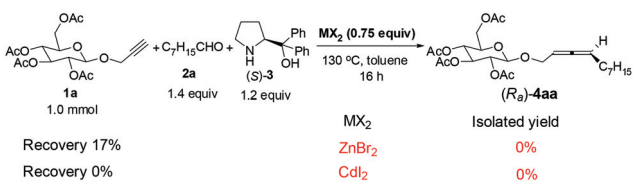


Fig. 1 Some typical natural or unnatural carbohydrates bearing an optically active allene unit with biopotential.



Scheme 1 Failed $ZnBr_2$- or CdI_2-mediated reaction of **1a** with **2a** in the presence of (*S*)-α,α-diphenylprolinol.

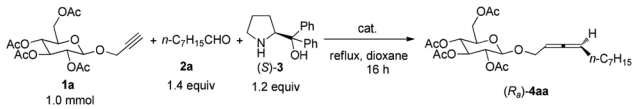
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†Electronic supplementary information (ESI) available: Experimental section, characterization of all compounds, and copies of ^1H and ^{13}C NMR spectra of selected compounds. CCDC 1013803. For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/c5qo00164a

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Initially, we conducted the reaction of **1a** with octanal **2a** (1.4 equiv.) and (*S*)-α,α-diphenylprolinol (*S*)-**3** (1.2 equiv.) under catalysis by $CuBr_2$ (40 mol%) in 1,4-dioxane under reflux. Fortunately, after 16 h, the reaction succeeded to afford the expected product (R_a)-**4aa** in 59% yield with 94% de (entry 1, Table 1). As a comparison, when CuI[10] was applied, the reaction afforded the expected product (R_a)-**4aa** in 67% yield, however, the diastereoselectivity is rather disappointing (29%) (entry 2, Table 1); compared with CuI, the reaction using CuBr gave (R_a)-**4aa** in the same yield, however, there is essentially no chiral induction for the allene axial chirality (entry 3, Table 1). The reaction in dioxane with $ZnBr_2$ still failed, indicating the beauty of the Cu^{2+} catalyst for such a complicated coordination environment of carbohydrates (entry 4, Table 1). Furthermore, it is interesting to observe that both the yield and de depend on the loading of the catalyst (entries 5–8, Table 1): when reducing the loading of $CuBr_2$ to 20 mol%, the diastereomeric excess was improved further to 98% (entry 5, Table 1). Control experiments show that the solvent effect is also striking: when we ran this reaction with $CuBr_2$ (20 mol%) in toluene, the diastereoselectivity is only 86% with a yield of 41% (entry 9, Table 1).

Table 1 Identifying a catalytic diastereoselective synthesis of (R_a)-**4aa**

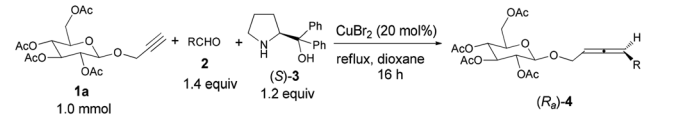


Entry	Cat. (mol%)	(R_a)-**4aa** Isolated yield (%)	de (%)
1	$CuBr_2$ (40)	59	94
2	CuI (40)	67	29
3	CuBr (40)	67	1
4	$ZnBr_2$ (40)	0[a]	—
5	**$CuBr_2$ (20)**	**49**	**98**
6	$CuBr_2$ (30)	53	96
7	$CuBr_2$ (50)	54	91
8	$CuBr_2$ (60)	57	88
9[b]	$CuBr_2$ (20)	41	86

[a] 46% of **1a** was recovered. [b] Toluene was used as the solvent.

The scope of the aldehydes has been demonstrated by using **1a** as the terminal alkyne with some of the typical results summarized in Table 1: both *n*-alkyl (entries 1–4, Table 2) and *sec*-alkyl (entries 5–6, Table 2) aldehydes, and even aromatic aldehydes **2g** and **2h** (at a much lower temperature) (entries 7–8, Table 2) may be used, affording the corresponding axially chiral allenyl glucopyranosides in decent yields with 97–99% de.

Table 2 The scope of aldehydes with 2-propynyl 2,3,4,6-tetra-*O*-acetyl-β-D-glucopyranoside **1a**



Entry	R	(R_a)-**4** Isolated yield (%)	de (%)
1	n-C_7H_{15} (**2a**)	49 ((R_a)-**4aa**)	98
2	Bn (**2b**)	55 ((R_a)-**4ab**)	97
3	$PhCH_2CH_2$ (**2c**)	50 ((R_a)-**4ac**)	99
4	*i*-Bu (**2d**)	54 ((R_a)-**4ad**)	97
5	Cy (**2e**)	57 ((R_a)-**4ae**)	99
6	*i*-Pr (**2f**)	53 ((R_a)-**4af**)	99
7[a]	*p*-ClC_6H_4 (**2g**)	54 ((R_a)-**4ag**)	99
8[a]	*p*-MeC_6H_4 (**2h**)	50 ((R_a)-**4ah**)	98

[a] The reaction was conducted at 70 °C for 24 h and 40 mol% of $CuBr_2$ was used.

The absolute configuration of the allene unit was established by the X-ray single crystal diffraction study of (R_a)-**4ae** with the glucopyranoside as the reference (Fig. 2).[11]

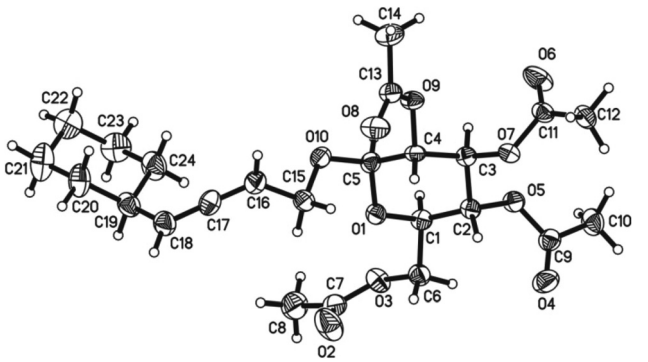
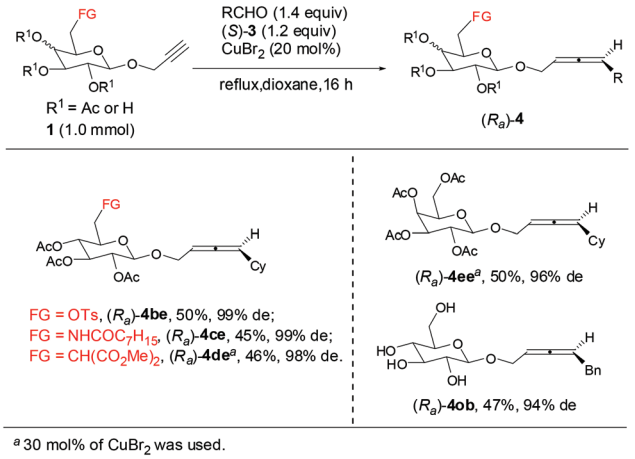


Fig. 2 ORTEP representation of (R_a)-**4ae**.

The reaction is also general with other carbohydrate-based terminal alkynes bearing different synthetically attractive functionalities without protection: for propargyl β-D-glucopyranosides **1b–1d**[12] bearing $-NHCOC_7H_{15}$, $-CH(CO_2Me)_2$, even –OTs unit at the *C*-6 position, the reactions all afforded the corresponding axially chiral allenyl β-D-glucopyranosides (R_a)-**4be–4de** in moderate yields with 98–99% de. When propargyl galactopyranoside **1e**[5] was used, the reaction afforded (R_a)-**4ee** in 50% yield with 96% de. It is worthy of mention that even the propargyl β-D-glucopyranoside **1o** with four free hydroxyl groups without protection could be converted to (R_a)-**4ob** in 47% yield with 94% de (Scheme 2).

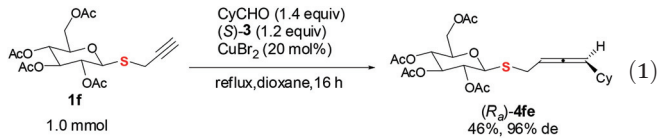
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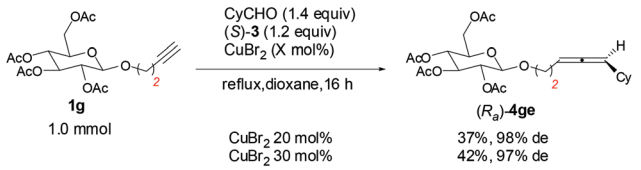
Scheme 2 The scope of propargyl glycosides.

Propargyl β-D-thioglucoside **1f**[13] could also be converted with 46% yield and 96% de to (R_a)-**4fe**, indicating that even the thio unit does not poison the Cu^{2+} catalyst (eqn (1)).



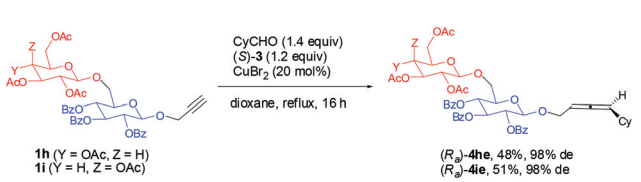
(1)

For homopropargyl β-D-glucopyranoside **1g**,[14] the expected product (R_a)-**4ge** was obtained in 37% yield with 98% de under catalysis with 20 mol% $CuBr_2$. Increasing the amount of $CuBr_2$ to 30 mol% could increase the yield to 42% with 97% de (Scheme 3).



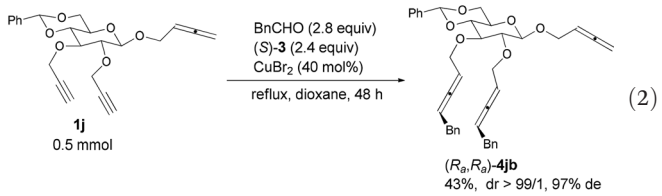
Scheme 3 Synthesis of axially chiral 3,4-allenyl β-D-glucopyranoside (R_a)-**4ge**.

This protocol is even suitable for the synthesis of more complex axially chiral allenyl disaccharides (R_a)-**4he** (48% yield, 98% de) and (R_a)-**4ie** (51% yield, 98% de) from **1h** and **1i**[15] (Scheme 4).



Scheme 4 Synthesis of axially chiral 2,3-allenyl disaccharides (R_a)-**4he** and (R_a)-**4ie**.

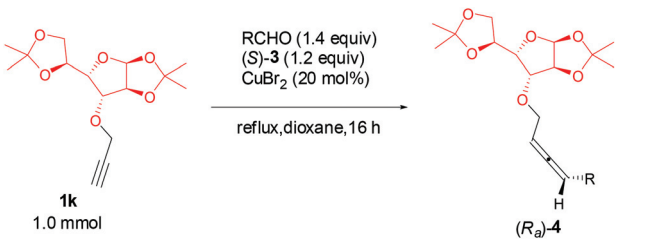
When we used the glucopyranoside **1j** bearing two terminal alkyne units at the *O*-2 and *O*-3 positions and a terminal allene unit, the reaction afforded the corresponding product (R_a,R_a)-**4jb** without touching the acetal and terminal allene units in 43% yield with 97% de (dr > 99/1), indicating that the axial allene chirality could be constructed at both α- and β-positions (eqn (2)).



(2)

Moreover, this method could be extended to construct axially chiral 2,3-allenyl five-membered furanosides in moderate yields with 96–99% de. No reaction occurred at the acetal unit (Table 3).

Table 3 The scope of aldehydes with propargyl furanoside **1k**



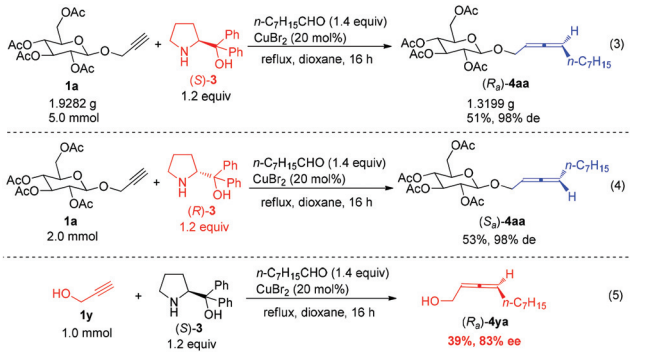
Entry	R	(R_a)-4 Isolated yield (%)	de (%)
1	*n*-C_7H_{15} (**2a**)	52 ((R_a)-**4ka**)	97
2	Cy (**2e**)	56 ((R_a)-**4ke**)	96
3[a]	*p*-MeC_6H_4 (**2h**)	49 ((R_a)-**4kh**)	98
4[a]	Ph (**2i**)	50 ((R_a)-**4ki**)	98
5[a]	*m*-ClC_6H_4 (**2j**)	46 ((R_a)-**4kj**)	99

[a] The reaction was conducted at 70 °C for 24 h and 40 mol% of $CuBr_2$ was used.

The reaction may be easily conducted on a gram scale showing its practicality (eqn (3), Scheme 5). The diastereoisomer (S_a)-**4aa** could be prepared with the prolinol enantiomer (*R*)-**3** with a similar level of de, indicating the absolute configuration of the allene axial chirality is determined by that of the amino alcohol **3**, not the skeleton of the carbohydrate (eqn (4), Scheme 5). However, it is interesting to observe that under the same conditions the reaction of propargyl alcohol **1y** afforded chiral 2,3-allenol (R_a)-**4ya** in only 39% yield with just 83% ee,[9a] indicating the unique role of the carbohydrate unit in both the yield and de – the sterically bulky carbohydrate

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unit ensures the stereoselectivity of the allene formation as a stereodictating protecting group (eqn (5), Scheme 5).[6b]



Scheme 5 Control experiments.

In conclusion, we have observed that $CuBr_2$ may catalyze the 1,3-disubstituted allene unit bearing a carbohydrate unit from aldehydes and carbohydrates bearing a terminal alkyne unit with very high diastereoselectivity. The following advances have been realized: (1) $CuBr_2$ has been identified as a catalyst compatible with the *in situ* generated water, the highly oxygenated nature of the carbohydrates, and the *in situ* generated imine; (2) the current method provides one-pot approach to chiral allenes with common functionalities such as hydroxyl, amide, malonate, even tosylate, enjoying a very wide scope. (3) The absolute configuration of the newly formed allene unit is determined by the absolute configuration of the readily available and cheap (*S*)- or (*R*)-α,α-diphenylprolinol,[16] while the carbohydrate unit acts as a stereodictating group to improve the diastereomeric excess. This one-pot highly diastereoselective approach for the construction of the chiral allene unit will greatly stimulate interest in the science of allenes and carbohydrates for organic chemists and medicinal chemists. Further studies including the synthetic application are being pursued in this laboratory.

Acknowledgements

Financial support from the National Natural Science Foundation of China (21232006) and National Natural Basic Research Program of China (2015CB856600) are greatly appreciated. We thank Mr Hongwen Luo in this group for reproducing the preparation of (R_a)-**4ac**, (R_a)-**4de**. Shengming Ma is a Qiu Shi Adjunct Professor at Zhejiang University.

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7 Periasamy *et al.* also reported the synthesis of axially chiral allenes from simple non-functionalized terminal alkynes

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by using α,α-diphenylprolinol right after our report (ref. 6*a*); we have not been able to reproduce their results in terms of ee and yield: the calibrated specific optical rotations of the same allenes with similar ee values from our study and the data in their report are different. The problem is most probably caused by inaccurate HPLC analysis for both the racemic and optically active allenes: M. Periasamy, N. Sanjeevakumar, M. Dalai, R. Gurubrahamam and P. O. Reddy, *Org. Lett.*, 2012, **14**, 2932. For details, see: ref. 6*c*. Later, we developed the $ZnBr_2$–CuBr-mediated protocol for simple terminal alkynes (see ref. 6*d*).

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